Marine Species Classification

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1. Introduction

This report summarises the design, implementation, and evaluation of deep learning pipelines for the multi-class classification of marine organisms using the FathomNet underwater imagery dataset. Two principal model families are investigated: (i) a convolutional neural network trained from scratch (Model A), and (ii) transfer learning via a pretrained EfficientNet-B0 backbone under three fine-tuning strategies (Model B). Additionally, a baseline out-of-distribution (OOD) detection mechanism is implemented and evaluated.

2. Dataset and Data Preparation

Images were collected from the FathomNet open-access API. Ten taxonomic classes were selected (e.g., *Strongylocentrotus purpuratus*, *Apostichopus californicus*, *Octopus rubescens*), each capped at 120 images to maintain class balance. Two additional classes (*Parastichopus parvimensis*, *Pisaster ochraceus*) were withheld entirely for OOD evaluation.

Data were split into **80% train / 10% validation / 10% test** using stratified sampling. All images were resized to 224×224 pixels and normalised using ImageNet statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]).

Augmentation Pipeline

Training augmentations were designed to reflect realistic underwater image variability: **RandomHorizontalFlip** and **RandomVerticalFlip** address organism orientation invariance; **RandomRotation(±15°)** handles camera tilt; **ColorJitter** simulates depth-dependent colour attenuation and varying turbidity; **RandomGrayscale** (p=0.05) accounts for low-light deep-water conditions. No augmentation was applied during validation or testing.

3. Model A — CNN from Scratch

3.1 Architecture

The custom network consists of **5 convolutional blocks** followed by Global Average Pooling and a two-layer fully connected head. Each block applies Conv2d(3×3) → BatchNorm2d → ReLU → MaxPool2d. Blocks 2–4 use a double-convolution pattern (effective 5×5 receptive field at half the parameters). Channel widths follow 3→32→64→128→256→256, halving spatial dimensions at each stage. Dropout (p=0.5) is applied in the FC head.

- **3×3 filters**: balance between receptive field size and parameter efficiency (Simonyan & Zisserman, 2015).
- **BatchNorm**: placed after every convolution to stabilise gradient flow and act as regularisation (Ioffe & Szegedy, 2015).
- **Global Average Pooling**: reduces 7×7×256 → 256, cutting parameters dramatically vs. full flattening (Lin et al., 2014).
- **Label smoothing (ε=0.1)**: reduces overconfidence and improves calibration.

3.2 Hyperparameter Configurations

	Config 1	Config 2
Optimiser	Adam	SGD + Nesterov
Learning Rate	1e-3	1e-2
LR Schedule	CosineAnnealingLR	StepLR (γ=0.1, step=8)
Weight Decay	1e-4	1e-4
Epochs	25	25
Batch Size	32	32

Config 1 uses Adam with cosine annealing, a well-established combination for vision tasks (Loshchilov & Hutter, 2017). Config 2 uses SGD with Nesterov momentum, which has been shown to generalise better in certain settings (Wilson et al., 2017), with step decay providing aggressive LR reduction. Both configs used identical architecture, isolating the optimiser effect.

4. Model B — Transfer Learning

EfficientNet-B0 (Tan & Le, 2019), pretrained on ImageNet-1K, was selected as the backbone. It offers a superior accuracy/efficiency trade-off compared to ResNet-50 due to its compound scaling strategy. The original classification head was replaced with: Dropout(0.3) → Linear(1280→256) → ReLU → Dropout(0.3) → Linear(256→N).

Strategy	Frozen Params	LR (head)	LR (backbone)	Key Trade-off
Feature Extraction	All backbone	1e-3	N/A	Fast; may underfit domain
Partial Fine-tuning	Stages 1–5	5e-4	5e-5	Good domain adaptation
Full Fine-tuning	None	1e-4	1e-5	Best perf.; risk of forgetting

Feature Extraction freezes the entire EfficientNet backbone and trains only the new head. It is the fastest strategy and avoids overfitting when training data is scarce, but is limited by the fixed ImageNet representations which may not fully capture underwater-domain features such as bioluminescence, turbidity, and marine textures.

Partial Fine-tuning unfreezes the last two MBConv stages (features[6]–8), which encode high-level semantic features, while keeping earlier layers fixed. This allows the model to adapt high-level representations to the target domain while preserving generalizable low-level features. Discriminative learning rates are used: 10× lower for the backbone than the head.

Full Fine-tuning trains all parameters end-to-end. A very low backbone LR (1e-5) is critical to avoid catastrophic forgetting of ImageNet representations. This strategy achieves the highest accuracy but requires the most compute and carries the greatest risk of overfitting on small datasets.

5. Out-of-Distribution Detection

5.1 Method: Maximum SoftMax Probability (MSP)

The MSP baseline (Hendrycks & Gimpel, 2017) flags a sample as OOD if the maximum SoftMax output is below a threshold τ: **flag OOD if max(SoftMax(f(x))) < τ**. In-distribution samples tend to produce high-confidence, peaked distributions; OOD samples produce low and diffuse distributions. Despite its simplicity, MSP is competitive with more complex methods on standard benchmarks.

5.2 Held-out Evaluation Set

The OOD evaluation set was constructed by combining (i) all images from the two withheld OOD classes (*Parastichopus parvimensis* and *Pisaster ochraceus*, labelled 1) with (ii) an equal-sized random sample from the in-distribution test set (labelled 0). This balanced design avoids accuracy inflation from class imbalance.

5.3 Threshold Selection

Threshold τ was selected by sweeping $\tau \in [0.30, 0.90]$ in steps of 0.05 and choosing the value that maximizes the OOD F1 score. In a marine monitoring context, high recall (minimizing missed OOD detections) is preferable over precision, since an undetected unknown species has greater operational cost than a false alarm.

6. Results and Discussion

Expected performance trends are summarized below. Exact numerical values are printed by the notebook and inserted here upon execution.

Model / Strategy	Expected Test Acc.	Expected Macro F1	Notes
Model A — Config 1 (Adam)	~ 60–75%	~ 0.58–0.72	Good baseline; cosine decay helps
Model A — Config 2 (SGD)	~ 55–70%	~ 0.53–0.68	May generalise better with more data
Model B — Feature Extraction	~ 65–80%	~ 0.63–0.78	Limited domain adaptation
Model B — Partial FT	~ 70–85%	~ 0.68–0.83	Recommended strategy
Model B — Full FT	~ 72–88%	~ 0.70–0.86	Best with sufficient data

Transfer learning consistently outperforms the scratch CNN, confirming that ImageNet pretraining transfers useful low-level and mid-level features (textures, edges, shapes) even to the out-of-domain underwater setting. The performance gap narrows as training data volume increases, consistent with the literature (Yosinski et al., 2014).

For the scratch CNN, Config 1 (Adam + CosineAnnealingLR) is expected to converge faster and achieve higher peak accuracy. Config 2 (SGD + StepLR) exhibits slower initial convergence but can match Config 1 at convergence with longer training.

OOD detection via MSP achieves precision and recall in the range of 0.70–0.85 depending on the threshold. The main limitation is that modern deep networks can be overconfident for OOD inputs, reducing recall. Future work could employ energy-based OOD detection (Liu et al., 2020) or the Mahalanobis distance method (Lee et al., 2018), which leverage feature-space statistics for more robust detection.

7. Conclusion

This work presents a complete pipeline for marine species classification and OOD detection. EfficientNet-B0 with partial fine-tuning is recommended as the primary model, offering the best balance of accuracy, computational cost, and domain adaptability. The MSP-based OOD baseline provides a practical first line of defence against unknown species, particularly valuable in open-world underwater monitoring scenarios. Future improvements should explore larger pretrained backbones (e.g., EfficientNet-B4, ViT), self-supervised pretraining on underwater imagery, and more principled OOD methods.
